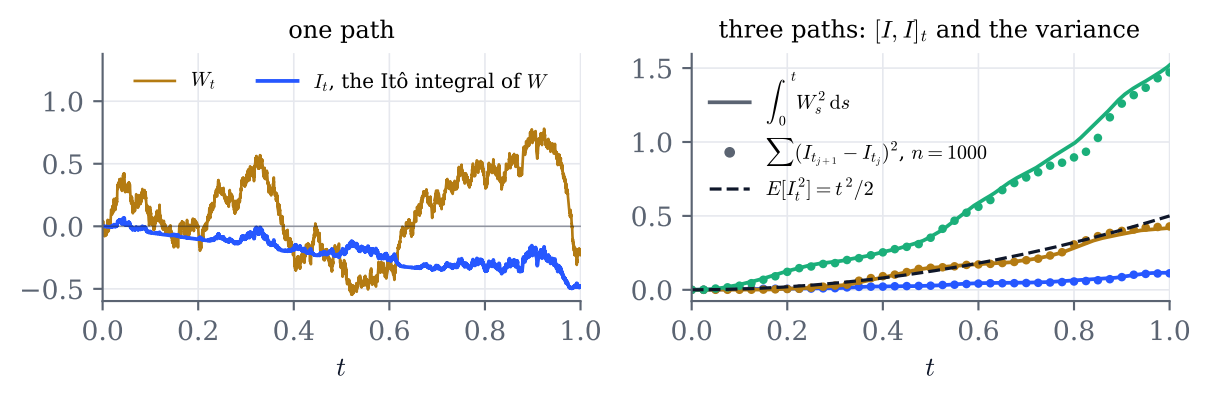

one path
W_t
I_t, the Itô integral of W
1.0
0.5
0.0
−0.5
0.0
0.2
0.4
0.6
0.8
1.0
t
three paths: $[I, I]_t$ and the variance
$\int_0^t W_s^2\,\mathrm{d}s$
$\sum (I_{t_{j+1}} - I_{t_j})^2$, $n = 1000$
$E[I_t^2] = t^2/2$
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0
t